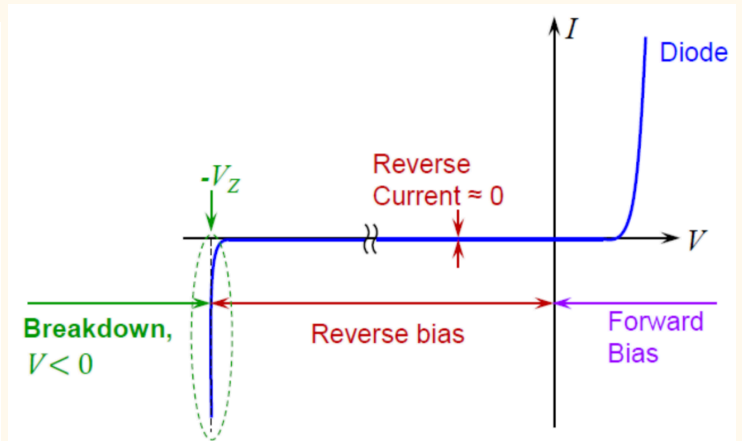
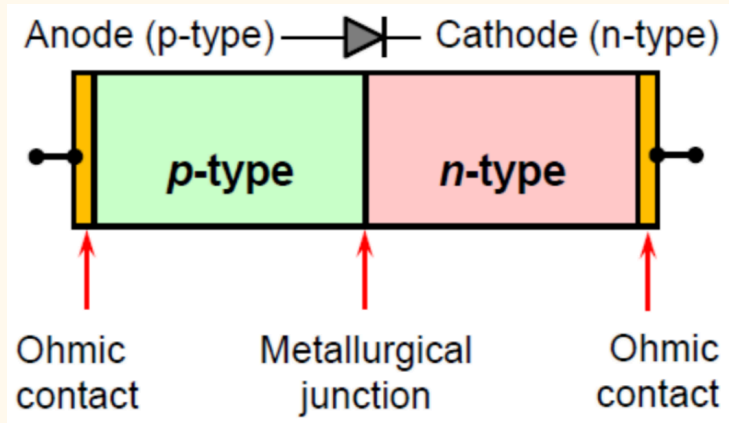


Diode Summary

Operating Regions and IV Characteristics



- CANNED APPLE
 - cathode is n
 - anode is p

IV Characteristics	
Substantially Forward Bias $V_D \gg V_{cut-in}$ or $I_D \gg I_S$	$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right) \approx I_S e^{\frac{V_D}{nV_T}}$ $V_D = nV_T \ln\left(\frac{I_D}{I_S}\right) = nV_T \ln(I_D) - nV_T \ln(I_S)$ • V_D and ln(I_D) has a linear relationship
Reverse Bias	$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right) \approx -I_S$
Reverse Breakdown	$V_D = V_Z$
Diode Parameters	
Thermal Voltage	$V_T = \frac{kT_{Kelvin}}{q} \quad V_T \approx 25 \text{ mV} @ T = 300 \text{ K} (298 \text{ K}) = 25^\circ\text{C}$ $k = \text{Boltzmann's constant} = 1.380649 \times 10^{-23} \text{ J/K}$ $q = \text{elementary charge} = 1.60217663 \times 10^{-19} \text{ C}$
Exponential/Ideality Factor	n 1 = ideal, reality 1 ~ 2
Reverse Saturation Current	I_S